

Unbiased exact diagonalization of quantum spin ice in the π -flux regime: a zero-transport projector for finite-size gauge contamination

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Abstract

The thermodynamics and dynamics of quantum spin ice (QSI) are governed by an emergent compact $U(1)$ gauge field. Its unfrustrated 0-*flux* sector ($J_{\pm} > 0$) is accessible to sign-problem-free quantum Monte Carlo (QMC); the frustrated π -*flux* sector ($J_{\pm} < 0$) realized in the dipolar-octupolar cerium pyrochlores carries a QMC sign problem, leaving exact diagonalization (ED) as the only unbiased finite-temperature method. ED, however, has a finite-size problem specific to QSI: a periodic cluster hosts short ice-preserving ring exchanges that wind the torus and have no infinite-lattice analogue. The dominant ones are four-site loops with coupling $g_4 = 4J_{\pm}^2/J_{zz}$, which is *second* order in $J_{\pm}$ and therefore parametrically larger than the genuine photon coupling $g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$; they contaminate exactly the emergent-gauge observables one wants (the low- T specific-heat peak and the low-energy $S^{zz}(\mathbf{q}, \omega)$). We remove this contamination with a *zero-transport projector*: retain only ice-manifold processes whose net emergent-charge transport $\boldsymbol{\delta} = \boldsymbol{\rho} + \mathbf{N} \cdot \mathbf{L}$ vanishes modulo the cluster periods. We prove, both mathematically and by direct computation, that the projector (i) annihilates every winding ring exchange at all orders in $J_{\pm}$, (ii) preserves every contractible ring exchange at full strength, and (iii) coincides with the operator-level twist average taken in the translationally-invariant gauge – clarifying why the naive twist averages used in practice fail (the boundary gauge removes only half; averaging observables rather than the Hamiltonian keeps the full spurious scale, gauge-invariantly). On the 16- and 32-site clusters the projector moves the low- T specific-heat peak and the low-energy S^{zz} weight from the artifact scale g_4 to the photon scale g_{hex} , for both flux signs. This makes a 32-site pyrochlore an unbiased laboratory for the π -flux gauge sector, to be benchmarked against 0-flux QMC and applied at the exchange parameters of the cerium pyrochlores.

1 The problem

Quantum spin ice and its emergent photon. The minimal model of QSI is the XXZ antiferromagnet on the pyrochlore lattice, written in the local $\langle 111 \rangle$ frames,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad 0 < |J_{\pm}| \ll J_{zz}. \quad (1)$$

For $J_{\pm} \rightarrow 0$ the ground manifold is the exponentially degenerate set of 2-in-2-out (ice-rule) states. A small $J_{\pm}$ lifts this degeneracy through virtual ring exchanges and generates the deconfined $U(1)$ lattice gauge theory of QSI: S^z is the emergent electric field, the third-order hexagonal ring exchange

$$g_{\text{hex}} = \frac{12|J_{\pm}|^3}{J_{zz}^2} \quad (2)$$

is the Maxwell term whose quantum = the emergent photon, and ice-rule defects are gapped, fractionalized spinons (pair threshold $\Delta_s \approx 0.8 J_{zz}$ on the clusters below). The *sign* of the ring coupling follows $\text{sgn}(J_\pm^3)$: $J_\pm > 0$ gives the unfrustrated 0-flux vacuum, $J_\pm < 0$ the frustrated π -flux vacuum. Thermodynamically the photon appears as a low-temperature Schottky peak in $C(T)$ at $T \sim g_{\text{hex}}$, far below the spinon (charge) peak at $T \sim J_{zz}$; dynamically it appears as low-energy spectral weight in the non-spin-flip channel $S^{zz}(\mathbf{q}, \omega)$.

Why π -flux needs exact diagonalization. In the S^z basis the off-diagonal matrix elements of eq. (1) are $-J_\pm$. For $J_\pm > 0$ they are negative and worldline/stochastic-series QMC is sign-problem-free: the 0-flux QSI is quantitatively characterized. For $J_\pm < 0$ the same elements are positive and QMC acquires a sign problem. The π -flux regime – believed to describe the cerium pyrochlores $\text{Ce}_2\text{Zr}_2\text{O}_7$, $\text{Ce}_2\text{Sn}_2\text{O}_7$, $\text{Ce}_2\text{Hf}_2\text{O}_7$ – is thus a QMC dead zone, and unbiased finite-temperature information must come from exact diagonalization of a finite cluster.

The finite-size problem. A periodic cluster is a flat torus, and its periodic identifications generate ice-preserving ring exchanges that *wind* the torus and do not exist on the infinite lattice. The shortest and most damaging are *four-site* loops. Because a four-site loop factorizes into two disjoint nearest-neighbour (NN) hops, it is generated at *second* order in $J_\pm$,

$$g_4 = \frac{4 J_\pm^2}{J_{zz}}, \quad \text{so} \quad \frac{g_4}{g_{\text{hex}}} = \frac{J_{zz}}{3|J_\pm|} \gg 1. \quad (3)$$

The artifact is thus *parametrically larger* than the physics, and it grows as $|J_\pm| \rightarrow 0$ – exactly the weak-coupling regime of the cerium materials. It lives entirely in the ice manifold, so it contaminates precisely the emergent-gauge observables: it pushes the low- T $C(T)$ peak and the low-energy S^{zz} weight up to the scale g_4 instead of g_{hex} (quantified in section 6). The spinon continuum (S^{+-} at $\omega \sim J_{zz}$) lies outside the ice manifold and is *not* contaminated.

Goal. Remove this finite-size contamination from the gauge sector of a periodic-cluster ED, rigorously and verifiably, so that the 32-site pyrochlore delivers unbiased π -flux thermodynamics and dynamical structure factor.

2 The effective gauge Hamiltonian and the winding loops

Cluster torus. The cluster carries N sites with periodic identifications generated by the rows of an integer matrix $\mathbf{L} \in \mathbb{Z}^{3 \times 3}$ (in units of the cubic lattice constant a); position space is the flat torus $T^3 = \mathbb{R}^3 / \mathbf{L}\mathbb{Z}^3$ with first homology $\mathbb{Z}^3$. Each NN bond (i, j) has a *wrap vector* $\mathbf{n}_{ij} = \text{round}[(\mathbf{r}_j - \mathbf{r}_i)\mathbf{L}^{-1}]$, zero in the interior and nonzero across the boundary.

Effective Hamiltonian. Degenerate $(J_\pm)$ -perturbation theory in the ice manifold yields $H_{\text{eff}} = \sum_{\mathcal{C}} c_{\mathcal{C}} O_{\mathcal{C}}$, a sum of ring-exchange operators $O_{\mathcal{C}} = (S_a^+ S_b^- S_c^+ S_d^- \dots)$ over ice-preserving cycles $\mathcal{C}$; because each intermediate resolvent is Ising ($\propto J_{zz}$), the coefficient $c_{\mathcal{C}}$ is exactly $\propto J_\pm^{|\mathcal{C}|/2}$. On the $2 \times 2 \times 2$ FCC cluster ($N = 32$) the ice manifold has dimension 2970, so H_{eff} is *fully diagonalizable* and its thermodynamics is numerically exact. Enumeration (Schrieffer–Wolff with per-process bookkeeping) gives the census in table 1: the winding four-loops and wrapping hexagons are the contamination; the contractible hexagons carry the photon.

Table 1: Census of ice-preserving ring exchanges on the two production clusters. “winding/wrapping” loops close only through the periodic boundary ($\mathbf{w} \neq \mathbf{0}$); “contractible” hexagons are genuine bulk plaquettes ($\mathbf{w} = \mathbf{0}$). The last column is the survival fraction under the zero-transport projector of section 3.

cluster	channel	count	order	kept by projector
$1 \times 1 \times 1$ cubic ($N=16$)	winding four-loops	36	$J_{\pm}^2$	0
	wrapping hexagons	48	$J_{\pm}^3$	0
	contractible hexagons	16	$J_{\pm}^3$	1
$2 \times 2 \times 2$ FCC ($N=32$)	winding four-loops	24	$J_{\pm}^2$	0
	wrapping hexagons	96	$J_{\pm}^3$	0
	contractible hexagons	32	$J_{\pm}^3$	1

3 The zero-transport projector

The gauge-invariant label that distinguishes a genuine (bulk) ring exchange from a torus-winding one is not homology but *charge transport*.

Definition 1 (Net charge transport). For an ice-manifold process taking configuration c to c' by raising a site set R and lowering a site set L through hops of total wrap $\mathbf{N} = \sum \mathbf{n}$, the net transport is

$$\boldsymbol{\delta} = \boldsymbol{\rho} + \mathbf{N} \cdot \mathbf{L}, \quad \boldsymbol{\rho} = \sum_{i \in R} \mathbf{r}_i - \sum_{j \in L} \mathbf{r}_j, \quad (4)$$

the physical displacement of the transported S^z (boson) density. $\boldsymbol{\rho}$ is the intrinsic dipole of the flipped-site set (home positions); $\mathbf{N} \cdot \mathbf{L} = \sum_k N_k \mathbf{L}_k$ is the wrap contribution.

Definition 2 (Zero-transport projector). $\mathcal{P}$ retains exactly the processes with $\boldsymbol{\delta} \equiv \mathbf{0} \pmod{\mathbf{L}}$, i.e. $\boldsymbol{\delta} \in \mathbf{L}\mathbb{Z}^3$, and deletes the rest.

Two implementations, cross-validated in section 5: (i) in perturbation theory, a row filter on the enumerated processes; (ii) nonperturbatively, a mask on the exactly downfolded ice-manifold Hamiltonian, keeping matrix element (c, c') iff its flip-set dipole $\boldsymbol{\rho} \equiv \mathbf{0} \pmod{\mathbf{L}}$. Because $\mathbf{N}\mathbf{L} \in \mathbf{L}\mathbb{Z}^3$ automatically, the criterion $\boldsymbol{\delta} \in \mathbf{L}\mathbb{Z}^3$ reduces to $\boldsymbol{\rho} \in \mathbf{L}\mathbb{Z}^3$: *the projector’s action on an operator depends only on its dipole, at every order.*

4 Proof that the projector works: mathematics

4.1 Ring exchanges are sums over virtual paths

The naive expectation is that a winding ring exchange picks up its homology holonomy $e^{-i\mathbf{w} \cdot \boldsymbol{\varphi}}$ under a boundary twist $\boldsymbol{\varphi}$ and so is removed by averaging over $\boldsymbol{\varphi}$. This is false, and the reason underlies everything.

Theorem 1 (Matched-pair factorization). *Let $\mathcal{C}$ be an ice-preserving simple cycle of length $2m$ with winding $\mathbf{w}$. At the leading order at which $O_{\mathcal{C}}$ appears, its coefficient in the boundary-gauge twisted $H_{\text{eff}}(\boldsymbol{\varphi})$ is*

$$c_{\mathcal{C}}(\boldsymbol{\varphi}) = -\frac{g_{\mathcal{C}}}{M} \sum_{\mu=1}^M e^{-i\mathbf{N}_{\mu} \cdot \boldsymbol{\varphi}}, \quad \sum_{\mu} \mathbf{N}_{\mu} = \mathbf{w} \quad (M=2), \quad (5)$$

where μ runs over the $M \geq 2$ perfect matchings of $\mathcal{C}$ into disjoint NN dimer moves and $\mathbf{N}_\mu$ is the total wrap of matching μ . For a two-matching loop the gauge-invariant magnitude is

$$|c_{\mathcal{C}}(\boldsymbol{\varphi})| = g_{\mathcal{C}} |\cos(\frac{1}{2} \mathbf{w} \cdot \boldsymbol{\varphi})|. \quad (6)$$

Proof. At leading order $O_{\mathcal{C}}$ is generated only by flipping each of its $2m$ spins once, i.e. by a product of m disjoint NN dimer moves – a perfect matching of the cycle; an even cycle has $M \geq 2$. Each hop $S_i^+ S_j^-$ carries $e^{-i \mathbf{n}_{ij} \cdot \boldsymbol{\varphi}}$, so matching μ carries $e^{-i \mathbf{N}_\mu \cdot \boldsymbol{\varphi}}$ with $\mathbf{N}_\mu = \sum_{\text{hops}} \mathbf{n}$. The M matchings share amplitude $g_{\mathcal{C}}/M$ (equal denominators and weight). For a 4-cycle $abcd$ the two matchings $\{ab, cd\}$ and $\{bc, da\}$ together traverse each bond once, so $\mathbf{N}_1 + \mathbf{N}_2 = \mathbf{w}$; (6) follows from $|1 + e^{-i \mathbf{w} \cdot \boldsymbol{\varphi}}| = 2 |\cos(\frac{1}{2} \mathbf{w} \cdot \boldsymbol{\varphi})|$. $\square$

The operator thus lives in *several* transport sectors at once; the holonomy is only the *relative* phase of the matchings, and the magnitude (6) *vanishes* at $\mathbf{w} \cdot \boldsymbol{\varphi} = \pi$ – which a single homology phase (constant magnitude) never does.

4.2 The projector removes winding, keeps contractible

Theorem 2 (Transport selection). *Let $\ell_0 = \min_{\mathbf{w} \neq \mathbf{0}} |\mathbf{w} \cdot \mathbf{L}|$ be the shortest nonzero cluster period. If every ring exchange under consideration has alternating dipole $|\boldsymbol{\rho}_{\mathcal{C}}| < \ell_0$, then $\mathcal{P}$ deletes every loop with $\boldsymbol{\rho}_{\mathcal{C}} \neq \mathbf{0}$ at all orders in $J_{\pm}$ and keeps every loop with $\boldsymbol{\rho}_{\mathcal{C}} = \mathbf{0}$ at full strength; on the production clusters these are exactly the winding and the contractible loops respectively.*

Proof. By the remark below eq. (4), $\mathcal{P}$ keeps $O_{\mathcal{C}}$ iff $\boldsymbol{\rho}_{\mathcal{C}} \in \mathbb{L}\mathbb{Z}^3$, at every order. Since $|\boldsymbol{\rho}_{\mathcal{C}}| < \ell_0$, the only lattice vector it can equal is $\mathbf{0}$: hence $\boldsymbol{\rho}_{\mathcal{C}} \in \mathbb{L}\mathbb{Z}^3 \iff \boldsymbol{\rho}_{\mathcal{C}} = \mathbf{0}$, and $\mathcal{P}$ keeps $O_{\mathcal{C}}$ iff $\boldsymbol{\rho}_{\mathcal{C}} = \mathbf{0}$.

A genuine bulk plaquette (regular [111]-hexagon, or its four-tetrahedron analogue) is centrosymmetric – its raised and lowered sites share a centroid – so $\boldsymbol{\rho} = \mathbf{0}$ and it is kept. A winding loop has $\boldsymbol{\rho} \neq \mathbf{0}$: were $\boldsymbol{\rho} = \mathbf{0}$, then in the two-matching factorization $\boldsymbol{\delta}_1 + \boldsymbol{\delta}_2 = \mathbf{w} \cdot \mathbf{L}$ with $|\mathbf{w} \cdot \mathbf{L}| \geq \ell_0$ would force some $|\boldsymbol{\delta}_\mu| \geq \frac{1}{2} \ell_0$, yet $\boldsymbol{\delta}_\mu = \boldsymbol{\rho} + \mathbf{N}_\mu \mathbf{L}$ with $\boldsymbol{\rho} = \mathbf{0}$ makes $\boldsymbol{\delta}_\mu$ a lattice vector, hence $\mathbf{0}$ or $\geq \ell_0$ – contradiction. So winding loops are deleted. $\square$

On the production clusters the hypothesis holds with room to spare: the alternating dipole of *every* four-loop and wrapping hexagon has the same magnitude $|\boldsymbol{\rho}| = a/\sqrt{2}$, below $\ell_0 = a$ (cubic) and $\sqrt{2}a$ (FCC), while every contractible hexagon has $\boldsymbol{\rho} = \mathbf{0}$ identically – giving the 0/0/1 survival pattern of table 1. The single residual bias is honest: a *contractible* process with a nonzero dipole ($\boldsymbol{\rho} \neq \mathbf{0}$, possible only for non-centrosymmetric loops at order $J_{\pm}^4$ and beyond) would also be removed; within the NN model these are $O(J_{\pm}^4/J_{zz}^3)$, parametrically below g_{hex} .

4.3 Relation to twist averaging: why the naive versions fail

Theorem 3 (Twist-average identity). *In the translationally-invariant (“smooth”) gauge – a uniform vector potential $\mathbf{A} = \mathbf{L}^{-1} \boldsymbol{\varphi}$ placing $\theta_{ij} = \mathbf{A} \cdot \mathbf{d}_{ij}$ on every bond – each process carries $e^{-i \mathbf{A} \cdot \boldsymbol{\delta}}$, and the operator-level twist average over the extended period is $\mathcal{P}$:*

$$\overline{H_{\text{eff}}}^{\text{smooth}} = \int H_{\text{eff}}(\boldsymbol{\varphi}) \frac{d\boldsymbol{\varphi}}{(\text{period})} = \mathcal{P} H_{\text{eff}}. \quad (7)$$

Proof. A hop $i \rightarrow j$ contributes $\mathbf{A} \cdot \mathbf{d}_{ij}$; summed along a process the phase is $\mathbf{A} \cdot (\sum \mathbf{d}) = \mathbf{A} \cdot \boldsymbol{\delta}$ with $\boldsymbol{\delta}$ its transport. Averaging $e^{-i \mathbf{A} \cdot \boldsymbol{\delta}} = e^{-i \boldsymbol{\varphi} \cdot (\mathbf{L}^{-1} \boldsymbol{\delta})}$ over the period on which the smooth-gauge H_{eff} is single-valued gives $\delta_{\mathbf{L}^{-1} \boldsymbol{\delta} \in \mathbb{Z}^3} = \delta_{\boldsymbol{\delta} \in \mathbb{L}\mathbb{Z}^3}$, i.e. $\mathcal{P}$. $\square$

So the projector *is* twist averaging, done correctly. This pins down the two ways the naive procedure fails, both verified numerically in section 5:

Corollary 1 (Failure modes). (a) Wrong gauge. *In the boundary-bond gauge, a winding loop’s zero-wrap matching ($\mathbf{N}_\mu = \mathbf{0}$) is phase-free, so the operator average keeps $\frac{1}{2}g_C$ rather than 0; the “half” is a gauge artifact.* (b) Wrong object. *Averaging an observable $\mathcal{O}(g)$ (a specific $C(T)$ or $S(\mathbf{q}, \omega)$) rather than the Hamiltonian keeps every even power of the gauge-invariant $|c_C(\boldsymbol{\varphi})| = g_C |\cos(\frac{1}{2}\mathbf{w} \cdot \boldsymbol{\varphi})|$, because $\overline{|c|^{2k}} \neq 0$; the spurious scale survives in any gauge. Since practice averages the measured quantity, naive twist averaging fails.*

5 Proof that the projector works: computation

Each mathematical claim is checked exactly on the 16-site cluster (where full ED is available) and, where relevant, on the 32-site FCC cluster.

(1) Matched-pair magnitude (6). For a winding four-loop [sites (0, 2, 8, 10), $\mathbf{w} = (-1, 0, -1)$] the exact second-order coefficient obeys, along $\mathbf{w}$,

$$|c(\boldsymbol{\varphi})|/g_4 = 1.000, 0.866, 0.707, 0.500, 0.000, \dots = |\cos(\frac{1}{2}\mathbf{w} \cdot \boldsymbol{\varphi})| \text{ to all digits,}$$

vanishing at $\mathbf{w} \cdot \boldsymbol{\varphi} = \pi$. A single homology phase would keep $|c| = g_4$ for all $\boldsymbol{\varphi}$; the vanishing is the two-matching interference of theorem 1.

(2) Transport selection (2). Direct enumeration confirms table 1: on both clusters *every* winding four-loop and wrapping hexagon has $\boldsymbol{\rho} \notin \mathbf{LZ}^3$ (deleted), while *every* contractible hexagon has $\boldsymbol{\rho} = \mathbf{0}$ (kept) – survival 0/0/1, to machine precision, at orders $J_\pm^2$ and $J_\pm^3$.

(3) Twist-average identity (3) and failure modes. The coefficient of the winding four-loop under the operator-level twist average is

$$-g_4 \text{ (bare)} \longrightarrow -\frac{1}{2}g_4 \text{ (boundary gauge)} \longrightarrow 0 \text{ (smooth gauge = } \mathcal{P}),$$

while a contractible hexagon is $+g_{\text{hex}}$ in all three – confirming both theorem 3 and corollary 1(a). For corollary 1(b), the corner-averaged *observable* keeps the spurious scale (section 6).

(4) Nonperturbative agreement. The dipole mask on the exactly downfolded 16-site Hamiltonian reproduces the perturbative $\mathcal{P}$: e.g. the low- T $C(T)$ peak at $J_\pm = -0.05$ agrees to 0.00122 vs. 0.00127 J_{zz} , validating the truncation.

6 Demonstration: the artifact removed, both flux sectors

Thermodynamics. Table 2 lists the low- T (gauge) peak of $C(T)$ in units of g_{hex} . Bare periodic ED places it at the artifact scale, drifting up as $g_4/g_{\text{hex}} = J_{zz}/3|J_\pm|$ toward weak coupling; the projector brings it to $O(g_{\text{hex}})$ for both flux signs. Corner-averaging the *observable* barely moves it (corollary 1(b)).

Table 2: Low- T specific-heat (gauge) peak $T_{\text{peak}}/g_{\text{hex}}$ on the 32-site FCC cluster (exact full spectrum of the 2970-state ice manifold, order $J_{\pm}^4$). Bare periodic ED sits at the artifact scale; $\mathcal{P}$ restores the photon scale $O(1)$.

$J_{\pm}/J_{zz}$	bare PBC	$\mathcal{P}$	g_4/g_{hex}
-0.03	5.4	0.6	11.1
-0.05	3.1	0.3	6.7
+0.046 (0-flux)	10.1	0.8	7.2

Table 3: Low- ω centroid of the emergent-photon structure factor S^{zz} (in units of g_4) at $J_{\pm} = -0.05 J_{zz}$. Twist averaging of the observable fails; the projector reaches the photon scale $\sim g_{\text{hex}} = 0.15 g_4$.

scheme	16-site	32-site FCC
bare PBC	$4.2 g_4$	$1.8 g_4$
twist-averaged observable (8 corners)	$2.6 g_4$	–
zero-transport projector $\mathcal{P}$	$0.47 g_4$	$0.28 g_4$

Dynamical structure factor. The emergent photon is the low-energy weight of the non-spin-flip channel $S^{zz}(\mathbf{q}, \omega) = \sum_n |\langle n | S_{\mathbf{q}}^z | 0 \rangle|^2 \delta(\omega - \omega_n)$; since S^z is diagonal in the ice basis this is computed exactly from the projected H_{eff} . Table 3 gives the low- ω weight centroid: bare ED places it at the artifact scale g_4 , the observable twist-average leaves it there, and only $\mathcal{P}$ brings it to the photon scale. (The spinon continuum S^{+-} at $\omega \sim J_{zz}$ is uncontaminated and is obtained separately by full-Hilbert-space methods.)

Both tables make the same point in the two experimentally relevant observables: the finite-size artifact sits at g_4 , twist averaging does not remove it, and $\mathcal{P}$ restores the photon scale g_{hex} for both 0- and π -flux.

7 Outlook: the π -flux campaign

The projector makes the 32-site pyrochlore an unbiased calculator of the QSI gauge sector, and the couplings scale as $J_{\pm}^k$ exactly, so a single effective-Hamiltonian build serves every coupling and both signs. The program is: (A) benchmark the projected 0-flux $C(T)$, $S(T)$ and photon S^{zz} against sign-problem-free QMC at a common coupling; (B) cross into π -flux and compute the same quantities where QMC cannot, resolving the emergent-photon lower peak; (C) evaluate at the measured exchange parameters of $\text{Ce}_2\text{Zr}_2\text{O}_7$, $\text{Ce}_2\text{Sn}_2\text{O}_7$ and $\text{Ce}_2\text{Hf}_2\text{O}_7$ and confront the neutron $S(\mathbf{q}, \omega)$ and the specific heat directly. The uncontaminated spinon continuum is supplied by full-Hilbert-space finite-temperature Lanczos, with twist averaging used for its legitimate purpose there – Brillouin-zone interpolation of the genuine bands, not contamination removal.